Course: Cryptography and Network Security
Code: CSN15403
Branch: B.Tech -5th Semester

Lecture – 10 : Primes and Related Congruence Equations

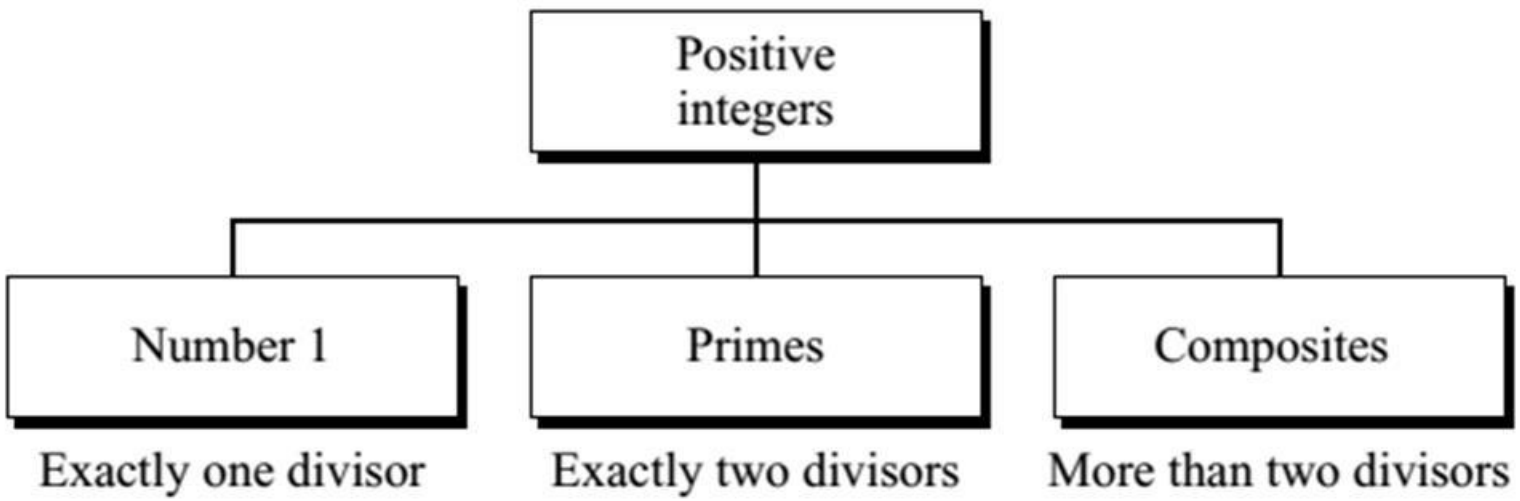
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PRIMES

- Asymmetric-key cryptography uses primes extensively.
- The topic of primes is a large part of any book on number theory.
- The positive integers can be divided into three groups: the number 1, primes, and composites.



PRIMES

- A positive integer is a prime if and only if it is exactly divisible by two integers, 1 and itself.
- A composite is a positive integer with more than two divisors.
- Two positive integers, a and b, are relatively prime, or coprime, if gcd (a, b) = 1.
- If p is a prime, then all integers 1 to p − 1 are relatively prime to p.
- Is there a finite number of primes or is the list infinite?
  - There is an infinite number of primes.
- A function called π(n) is defined that finds the number of primes smaller than or equal to n.

$$\pi(1) = 0 \quad \pi(2) = 1 \quad \pi(3) = 2 \quad \pi(10) = 4 \quad \pi(20) = 8 \quad \pi(50) = 15 \quad \pi(100) = 25$$

PRIMES

- But if n is very large, how can we calculate π(n)?
- The answer is that we can only use approximation.

$$[n / (\ln n)] \quad < \quad \pi(n) \quad < \quad [n/(\ln n - 1.08366)]$$

- Gauss discovered the upper limit; Lagrange discovered the lower limit.
- Find the number of primes less than 1,000,000.
- The approximation gives the range 72,383 to 78,543. The actual number of primes is 78,498.

Checking for Primeness

- Given a number n, how can we determine if n is a prime?
- The answer is that we need to see if the number is divisible by all primes less than √n.
- Is 97 a prime?
  - The floor of √97= 9.
  - The primes less than 9 are 2, 3, 5, and 7.
  - We need to see if 97 is divisible by any of these numbers.
  - It is not, so 97 is a prime.
- Is 301 a prime?
  - The floor of √301= 17.
  - We need to check 2, 3, 5, 7, 11, 13, and 17.
  - The numbers 2, 3, and 5 do not divide 301, but 7 does.
  - Therefore 301 is not a prime.

Euler's Phi-Function

- Euler's phi-function, $\phi(n)$, which is sometimes called the Euler's totient function plays a very important role in cryptography.
- The function finds the number of integers that are both smaller than n and relatively prime to n.
- The following helps to find the value of$\phi(n)$

1. $\phi(1) = 0$.
2. $\phi(p) = p - 1$ if p is a prime.
3. $\phi(m \times n) = \phi(m) \times \phi(n)$ if m and n are relatively prime.
4. $\phi(p^e) = p^e - p^{e-1}$ if p is a prime.

Euler's Phi-Function

- We can combine the above four rules to find the value of $\phi(n)$. For example, if n can be factored as

$$n = p_1^{e1} \times p_2^{e2} \times \ldots \times p_k^{ek}$$

- Then we combine the third and the fourth rule to find

$$\phi(n) = (p_1^{e_1} - p_1^{e_1-1}) \times (p_2^{e_2} - p_2^{e_2-1}) \times \cdots \times (p_k^{e_k} - p_k^{e_k-1})$$

The difficulty of finding $\phi(n)$ depends on the difficulty of finding the factorization of n.

Euler's Phi-Function

- Example 1
  - What is the value of $\phi(13)$?
- Solution
  - Because 13 is a prime, $\phi(13) = (13 - 1) = 12$.
- Example 2
  - What is the value of $\phi(10)$?
- Solution
  - We can use the third rule: $\phi(10) = \phi(2) \times \phi(5) = 1 \times 4 = 4$, because 2 and 5 are primes.

Euler's Phi-Function

- Example 3
  - What is the value of $\phi(240)$?
- Solution
  - We can write $240 = 2^4 \times 3^1 \times 5^1$. Then

$$\phi(240) = (2^4 - 2^3) \times (3^1 - 3^0) \times (5^1 - 5^0) = 64$$

- Example 4
  - Can we say that $\phi(49) = \phi(7) \times \phi(7) = 6 \times 6 = 36$????

Euler's Phi-Function

- Example 3
  - What is the value of $\phi(240)$?
- Solution
  - We can write $240 = 2^4 \times 3^1 \times 5^1$. Then

    $$\phi(240) = (2^4 - 2^3) \times (3^1 - 3^0) \times (5^1 - 5^0) = 64$$
- Example 4
  - Can we say that $\phi(49) = \phi(7) \times \phi(7) = 6 \times 6 = 36$????
- Solution
  - No. The third rule applies when *m* and *n* are relatively prime. Here $49 = 7^2$. We need to use the fourth rule: $\phi(49) = 7^2 - 7^1 = 42$.

Euler's Phi-Function

- Example 5
  - What is the number of elements in Z_{14}^*?
- Solution
  - The answer is $\phi(14) = \phi(7) \times \phi(2) = 6 \times 1 = 6$. The members are 1, 3, 5, 9, 11, and 13.

Interesting point: If n > 2, the value of $\phi(n)$ is even.

Fermat's Little Theorem

- First Version
  - If p is a prime and a is an integer such that p does not divide a,

$$a^{p-1} \equiv 1 \bmod p$$

- Second Version
  - Removes the condition on a
  - If p is prime and a is an integer,

$$a^{p} \equiv a \bmod p$$

Fermat's Little Theorem

- Example 1
  - Find the result of $6^{10} \bmod 11$.
- Solution
  - We have $6^{10} \bmod 11 = 1$. This is the first version of Fermat's little theorem where $p = 11$.
- Example 2
  - Find the result of $3^{12} \bmod 11$.
- Solution

Fermat's Little Theorem

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- Example 2
  - Find the result of $3^{12} \bmod 11$.
- Solution
  - Here the exponent (12) and the modulus (11) are not the same. With substitution this can be solved using Fermat's little theorem.

$$3^{12} \bmod 11 = (3^{11} \times 3) \bmod 11 = (3^{11} \bmod 11)\,(3 \bmod 11) = (3 \times 3) \bmod 11 = 9$$

Fermat's Little Theorem

- Multiplicative Inverses

$$a^{-1} \bmod p = a^{p-2} \bmod p$$

- The answers to multiplicative inverses modulo a prime can be found without using the extended Euclidean algorithm:

a. $8^{-1} \bmod 17 = 8^{17-2} \bmod 17 = 8^{15} \bmod 17 = 15 \bmod 17$

b. $5^{-1} \bmod 23 = 5^{23-2} \bmod 23 = 5^{21} \bmod 23 = 14 \bmod 23$

c. $60^{-1} \bmod 101 = 60^{101-2} \bmod 101 = 60^{99} \bmod 101 = 32 \bmod 101$

d. $22^{-1} \bmod 211 = 22^{211-2} \bmod 211 = 22^{209} \bmod 211 = 48 \bmod 211$

Euler's Theorem

- The modulus in the Fermat theorem is a prime, the modulus in Euler's theorem is an integer.
  - First Version
    - If a and n are coprime,

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

  - Second Version
    - Removes the condition that a and n should be coprime

$$a^{k \times \phi(n) + 1} \equiv a \pmod{n}$$

The second version of Euler's theorem is used in the RSA cryptosystem

Euler's Theorem

- Example 1
  - Find the result of $6^{24} \bmod 35$.
- Solution
  - We have $6^{24} \bmod 35 = 6^{\phi(35)} \bmod 35 = 1$.
- Example 2
  - Find the result of $20^{62} \bmod 77$???

Euler's Theorem

- Example 1
  - Find the result of $6^{24} \bmod 35$.
- Solution
  - We have $6^{24} \bmod 35 = 6^{\phi(35)} \bmod 35 = 1$.
- Example 2
  - Find the result of $20^{62} \bmod 77$.
- Solution

If we let $k = 1$ on the second version, we have

$$20^{62} \bmod 77 = (20 \bmod 77)\,(20^{\phi(77)+1} \bmod 77) \bmod 77$$
$$= (20)(20) \bmod 77 = 15.$$

Euler's Theorem

- Multiplicative Inverses
  - Euler's theorem can be used to find multiplicative inverses modulo a composite.

$$a^{-1} \bmod n = a^{\phi(n)-1} \bmod n$$

Euler's Theorem

- Example
  - The answers to multiplicative inverses modulo a composite can be found without using the extended Euclidean algorithm if we know the factorization of the composite:

a. $8^{-1} \bmod 77 = 8^{\phi(77)-1} \bmod 77 = 8^{59} \bmod 77 = 29 \bmod 77$

b. $7^{-1} \bmod 15 = 7^{\phi(15)-1} \bmod 15 = 7^{7} \bmod 15 = 13 \bmod 15$

c. $60^{-1} \bmod 187 = 60^{\phi(187)-1} \bmod 187 = 60^{159} \bmod 187 = 53 \bmod 187$

d. $71^{-1} \bmod 100 = 71^{\phi(100)-1} \bmod 100 = 71^{39} \bmod 100 = 31 \bmod 100$

Generating Primes

Mersenne Primes $\mathbf{M_p = 2^p - 1}$

$M_2 = 2^2 - 1 = 3$
$M_3 = 2^3 - 1 = 7$
$M_5 = 2^5 - 1 = 31$
$M_7 = 2^7 - 1 = 127$
$M_{11} = 2^{11} - 1 = 2047$ **Not a prime ($2047 = 23 \times 89$)**
$M_{13} = 2^{13} - 1 = 8191$
$M_{17} = 2^{17} - 1 = 131071$

A number in the form $M_p = 2^p - 1$ is called a Mersenne number and may or may not be a prime.

Generating Primes

Fermat Primes $F_n = 2^{2^n} + 1$

$F_0 = 3$
$F_1 = 5$
$F_2 = 17$
$F_3 = 257$
$F_4 = 65537$
$F_5 = 4294967297 = 641 \times 6700417$ **Not a prime**

No number greater than F4 has been proven to be a prime.
As a matter of fact many numbers up to F24 have been proven to be composite numbers.

Primality Testing

- Finding an algorithm to correctly and efficiently test a very large integer and output a prime or a composite has always been a challenge in number theory.
- Two types
  - Deterministic Algorithms
  - Probabilistic Algorithms

Primality Testing

```
Divisibility_Test (n)                    // n is the number to test for primality
{
  r ← 2
  while (r < √n )
  {
   if (r | n) return "a composite"
   r ← r + 1
  }
  return "a prime"
}
```

The bit-operation complexity of the divisibility test is $O(2^{n/2})$ (exponential)

Deterministic Algorithms

```
Divisibility_Test (n)                    // n is the number to test for primality
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  }
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}
```

The bit-operation complexity of the divisibility test is $O(2^{n/2})$ (exponential)

Deterministic Algorithms

- Example
  - Assume n has 200 bits. What is the number of bit operations needed to run the divisibility-test algorithm?
- Solution
  - The bit-operation complexity of this algorithm is $2^{n_b/2}$. This means that the algorithm needs 2^{100} bit operations. On a computer capable of doing 2^{30} bit operations per second, the algorithm needs 2^{70} seconds to do the testing !!!!!

Deterministic Algorithms

AKS Algorithm

$$(x-a)^p \equiv (x^p - a) \bmod p.$$

$$O((\log_2 n_b)^{12})$$

Tutorial

- Example
  - Assume n has 200 bits. What is the number of bit operations needed to run the AKS algorithm?
- Solution
  - This algorithm needs only $(\log_2 200)^{12}$ = 39,547,615,483 bit operations. On a computer capable of doing 1 billion bit operations per second, the algorithm needs only 40 seconds.

Probabilistic Algorithms

Fermat Test

If n is a prime, then $a^{n-1} \equiv 1 \bmod n$.

If n is a prime, $a^{n-1} \equiv 1 \bmod n$
If n is a composite, it is possible that $a^{n-1} \equiv 1 \bmod n$

Probabilistic Algorithms

- Example
  - Does the number 561 pass the Fermat test?
- Solution
  - Use base 2

$$2^{561-1} = 1 \bmod 561$$

  - The number passes the Fermat test, but it is not a prime, because 561 = 33 × 17.

Probabilistic Algorithms

Square Root Test

If n is a prime, $\sqrt{1} \bmod n = \pm 1$.
If n is a composite, $\sqrt{1} \bmod n = \pm 1$ and possibly other values.

- Example
  - What are the square roots of 1 mod n if n is 7 (a prime)?
- Solution
  - The only square roots are 1 and −1. We can see that

$$1^2 = 1 \bmod 7 \qquad (-1)^2 = 1 \bmod 7$$
$$2^2 = 4 \bmod 7 \qquad (-2)^2 = 4 \bmod 7$$
$$3^2 = 2 \bmod 7 \qquad (-3)^2 = 2 \bmod 7$$

  - Note that we don't have to test 4, 5 and 6 because $4 = -3 \bmod 7$, $5 = -2 \bmod 7$ and $6 = -1 \bmod 7$.

Probabilistic Algorithms

Square Root Test

- Example
  - What are the square roots of 1 mod n if n is 8 (a composite)?
- Solution
  - There are four solutions: 1, 3, 5, and 7 (which is −1). We can see that

$$1^2 = 1 \bmod 8 \qquad (-1)^2 = 1 \bmod 8$$
$$3^2 = 1 \bmod 8 \qquad 5^2 = 1 \bmod 8$$

Probabilistic Algorithms

Square Root Test

- Example
  - What are the square roots of 1 mod n if n is 17 (a prime)?
- Solution
  - There are only two solutions: 1 and −1

$1^2 = 1 \bmod 17$	$(-1)^2 = 1 \bmod 17$
$2^2 = 4 \bmod 17$	$(-2)^2 = 4 \bmod 17$
$3^2 = 9 \bmod 17$	$(-3)^2 = 9 \bmod 17$
$4^2 = 16 \bmod 17$	$(-4)^2 = 16 \bmod 17$
$5^2 = 8 \bmod 17$	$(-5)^2 = 8 \bmod 17$
$6^2 = 2 \bmod 17$	$(-6)^2 = 2 \bmod 17$
$(7)^2 = 15 \bmod 17$	$(-7)^2 = 15 \bmod 17$
$(8)^2 = 13 \bmod 17$	$(-8)^2 = 13 \bmod 17$

Probabilistic Algorithms

Square Root Test

- Example
  - What are the square roots of 1 mod n if n is 22 (a composite)??????

- Solution
  - Surprisingly, there are only two solutions, +1 and −1, although 22 is a composite.

$$1^2 = 1 \bmod 22$$
$$(-1)^2 = 1 \bmod 22$$

Probabilistic Algorithms

- Miller-Rabin Test

$$n - 1 = m \times 2^k$$

$$a^{n-1} = a^{m \times 2^k} = [a^m]^{2^k} = [a^m]^{2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}} \quad (k \text{ times})$$

The Miller-Rabin test needs from step 0 to step k − 1.

Probabilistic Algorithms

```
Miller_Rabin_Test (n, a)                    // n is the number; a is the base.
{
    Find m and k such that n − 1 = m × 2^k
    T ← a^m mod n
    if (T = ± 1)   return "a prime"
    for (i ← 1 to k − 1)                    // k − 1 is the maximum number of steps.
    {
        T ← T^2 mod n
        if (T = +1) return "a composite"
        if (T = −1) return "a prime"
    }
    return "a composite"
}
```

Probabilistic Algorithms

- Example
  - Does the number 561 pass the Miller-Rabin test?
- Solution
  - Using base 2, let $561 - 1 = 35 \times 2^4$, which means m = 35, k = 4, and a = 2.

Initialization:	$T = 2^{35} \bmod 561 = 263 \bmod 561$	
$k = 1$:	$T = 263^2 \bmod 561 = 166 \bmod 561$	
$k = 2$:	$T = 166^2 \bmod 561 = 67 \bmod 561$	
$k = 3$:	$T = 67^2 \bmod 561 = +1 \bmod 561$	$\rightarrow$ **a composite**

Probabilistic Algorithms

- Example
  - We already know that 27 is not a prime. Let us apply the Miller-Rabin test.
- Solution
  - With base 2, let $27 - 1 = 13 \times 2^1$, which means that $m = 13$, $k = 1$, and $a = 2$. The initialization step: $T = 2^{13} \bmod 27 = 11 \bmod 27$. However, because the algorithm enters the loop only once, it returns a composite.

Probabilistic Algorithms

- Example
  - We know that 61 is a prime, let us see if it passes the Miller-Rabin test.
- Solution
  - We use base 2.

$61 - 1 = 15 \times 2^2 \rightarrow m = 15 \quad k = 2 \quad a = 2$

Initialization: $T = 2^{15} \bmod 61 = 11 \bmod 61$

$k = 1$ $T = 11^2 \bmod 61 = -1 \bmod 61 \rightarrow$ **a prime**

Recommended Primality test

- Combination of the divisibility test and the Miller-Rabin test.
- Example
  - The number 4033 is a composite (37 × 109). Does it pass the recommended primality test?
- Solution
  1. Perform the divisibility tests first. The numbers 2, 3, 5, 7, 11, 17, and 23 are not divisors of 4033.
  2. Perform the Miller-Rabin test with a base of 2, $4033 - 1 = 63 \times 64$, which means m is 63 and k is 6.

Initialization: $T \equiv 2^{63} \pmod{4033} \equiv 3521 \pmod{4033}$

$k = 1$ $\quad T \equiv T^2 \equiv 3521^2 \pmod{4033} \equiv -1 \pmod{4033}$ $\rightarrow$ **Passes**

Recommended Primality test

But we are not satisfied. We continue with another base, 3.

Initialization: $T \equiv 3^{63} \pmod{4033} \equiv 3551 \pmod{4033}$

$k = 1$	$T \equiv T^2 \equiv 3551^2 \pmod{4033} \equiv 2443 \pmod{4033}$
$k = 2$	$T \equiv T^2 \equiv 2443^2 \pmod{4033} \equiv 3442 \pmod{4033}$
$k = 3$	$T \equiv T^2 \equiv 3442^2 \pmod{4033} \equiv 2443 \pmod{4033}$
$k = 4$	$T \equiv T^2 \equiv 2443^2 \pmod{4033} \equiv 3442 \pmod{4033}$
$k = 5$	$T \equiv T^2 \equiv 3442^2 \pmod{4033} \equiv 2443 \pmod{4033}$ **→Failed (composite)**